TESS planet candidate around TIC 452464529

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Summary

Signal found by the bls search of a public-data TESS pipeline (SPOC 120 s PDCSAP light curves, sectors 64, 65, 66, 93). Final tier: **candidate: likely planet**. Catalogue cross-match (TOIs, CTOIs, published planets, SPOC TCEs, TESS eclipsing binaries): **TOI 3025.01** (TOI). CNN score (calibrated): 0.916. TRICERATOPS (10 Monte Carlo runs, mean ± error of the mean): FPP $= 0.0174 \pm 0.0039$, NFPP $= 6.2e-11$.

Host star (TIC v8)

RA, Dec (deg)	139.04489, -76.505841
TESS magnitude	13.5
Radius ($R_\odot$)	1.42 ± –
$T_{\rm eff}$ (K)	5823 ± 122
Density ($\rho_\odot$)	0.365 ± –
Distance (pc)	889

Transit fit

Limb-darkened model (quadratic law, Kipping q_1, q_2; GPU float64, 1024 annuli, 1024x4096 tables, 15x supersampling), emcee with 48 walkers, 8000 steps (430 burn-in), run until the chain exceeded 50 autocorrelation times. Medians and 68% intervals.

Mid-transit time T_0 ($\mathrm{BJD_{TDB}}$)	$2460092.52528^{+0.00070}_{-0.00070}$
Period P (days)	$3.6795026^{+0.0000073}_{-0.0000073}$
Radius ratio $R_p/R_\star$	$0.0896^{+0.0011}_{-0.0011}$
Impact parameter b	$0.18^{+0.15}_{-0.12}$
Stellar density $\rho_\star$ (g cm^{-3})	$0.495^{+0.029}_{-0.055}$
Limb darkening q_1	$0.039^{+0.057}_{-0.027}$
Limb darkening q_2	$0.37^{+0.35}_{-0.25}$
Depth (ppm)	8497^{+206}_{-195}
Duration T_{14} (hours)	$4.283^{+0.049}_{-0.045}$
$a/R_\star$	$7.07^{+0.14}_{-0.27}$
Inclination (deg)	$88.5^{+1.0}_{-1.3}$
Planet radius ($R_\oplus$)	$13.91^{+1.00}_{-0.99}$
Semi-major axis (au)	$0.0464^{+0.0035}_{-0.0036}$
Equilibrium temperature (K, $A = 0$)	1553^{+43}_{-38}
Insolation ($S_\oplus$)	968^{+110}_{-91}

Vetting

Test	Value	Pass condition	Result
Transit SNR (correlated noise)	48.8	≥ 7.1	pass
Transits observed	26	≥ 2 (periodic)	pass
Odd/even depth difference (σ)	0.0263	≤ 4	pass
Secondary eclipse (σ)	1.69	≤ 3.5	pass
Ingress fraction (V shape)	0.109	≤ 0.4	pass
Planet radius ($R_\oplus$)	14.3	≤ 25	pass
Centroid source offset (arcsec)	1.43	4σ and $>21''$	pass
Difference image offset (arcsec)	2.89	4σ and $>10.5''$	pass
Difference image significance (σ)	1.15	–	–
CCD common mode (σ)	0.143	≤ 4	pass
Crowding (CROWDSAP)	0.86	≥ 0.8	pass
Stellar density ratio (max/TIC)	0.967	≥ 0.2	pass

Caveats

- This is a candidate with statistical evidence, not a confirmed planet: confirmation needs high-resolution imaging, reconnaissance spectroscopy and radial velocities.
- TRICERATOPS was run without a high-resolution imaging contrast curve, so close bound companions are not excluded.
- The MCMC converged (chain longer than 50 autocorrelation times, stable estimate) (acceptance 0.23, 8000 steps, about 4214 independent samples); errors are scaled by the time-averaging factor $\beta = 1.00$ for correlated noise.
- Stellar radius and effective temperature come from TIC v8; if the TIC gives no error, 7% (radius) and 150 K were assumed.

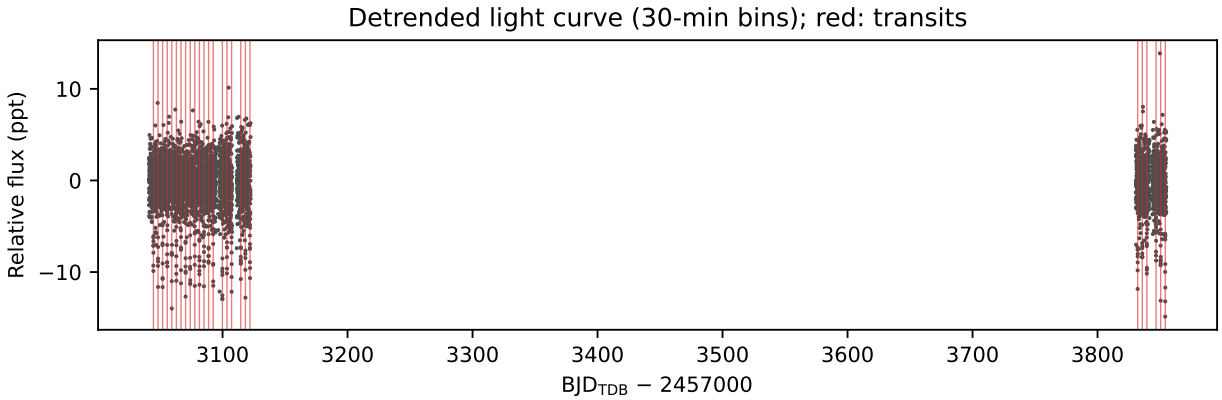


Figure 1: Detrended SPOC PDCSAP light curve (transit-protected biweight).

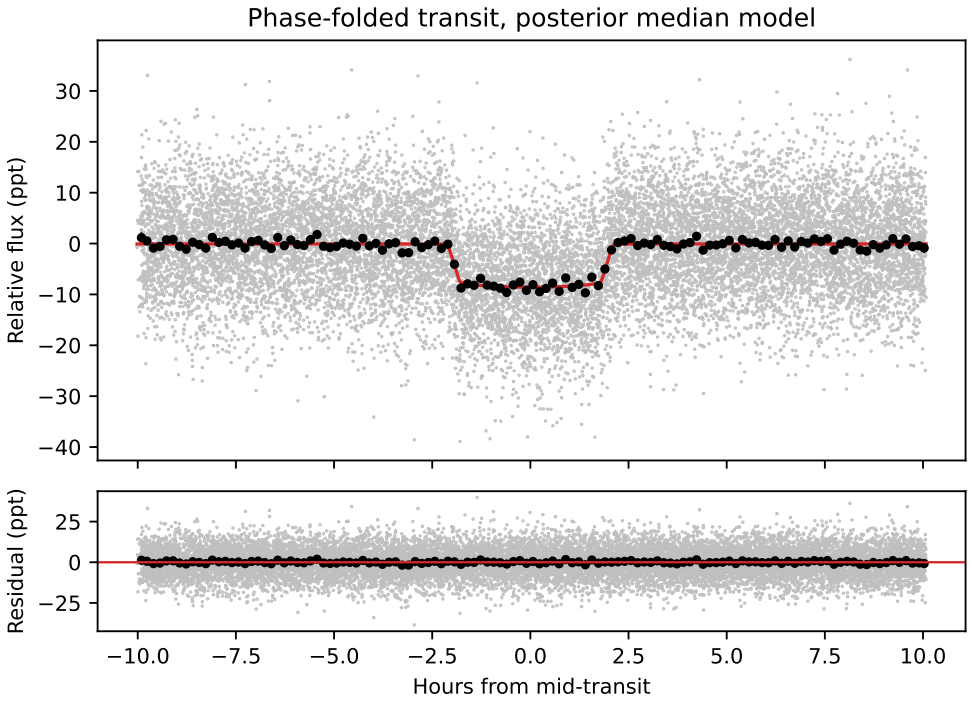


Figure 2: Phase-folded transit with the posterior median limb-darkened model.

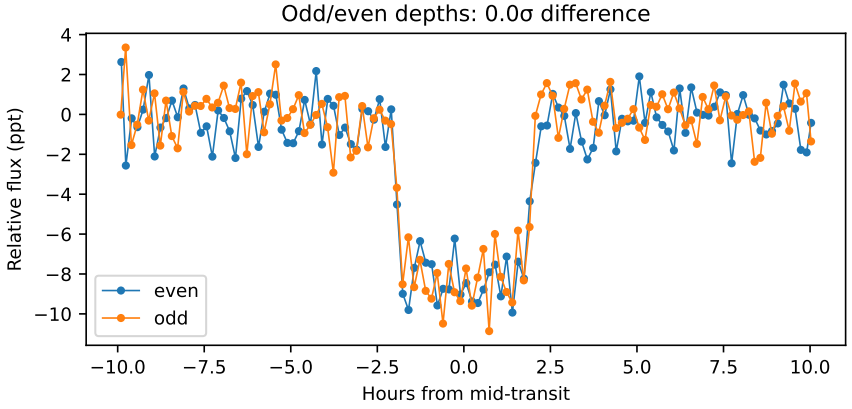


Figure 3: Odd and even transits, binned to 10 minutes.

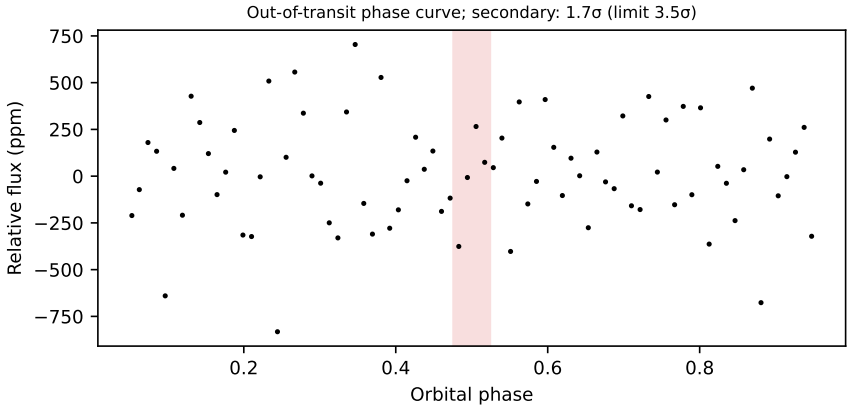


Figure 4: Phase curve; the shaded band is the secondary eclipse window for a circular orbit.

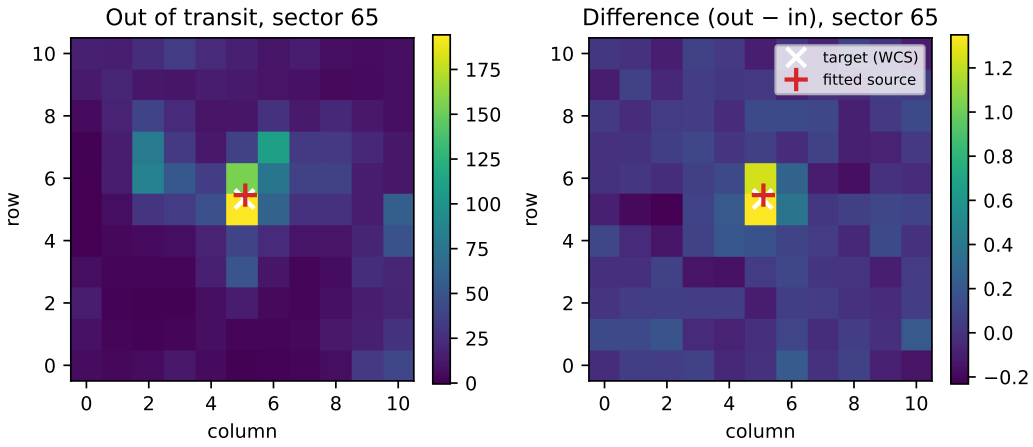


Figure 5: SPOC target pixel file: out-of-transit image and difference image. White cross: target catalogue position; red plus: fitted source.

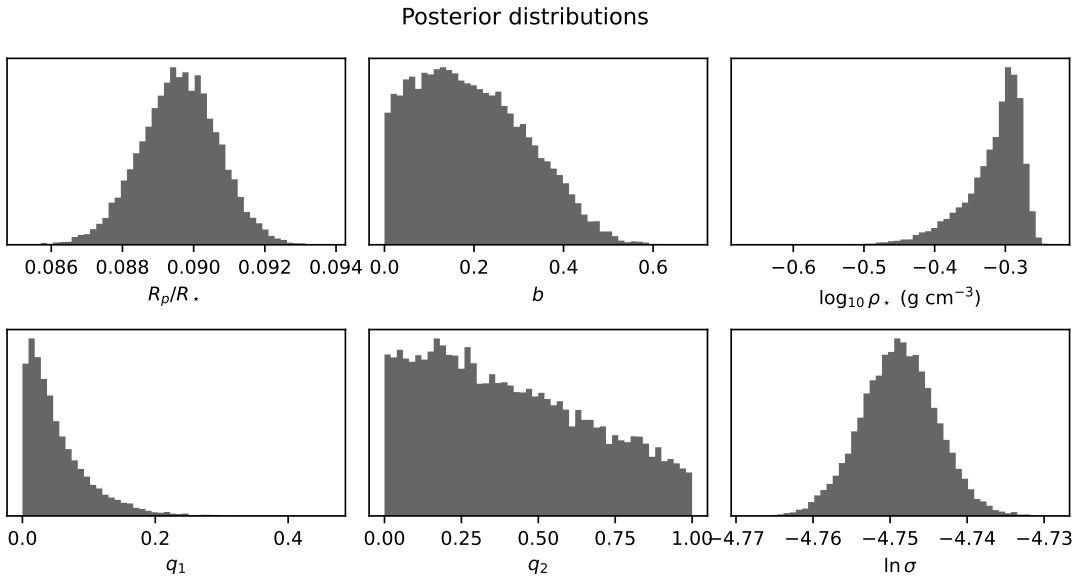


Figure 6: Posterior distributions of the fitted parameters.